Chapter 1. Set Theory and Logic

Sets and functions, relations, countability, recursive definition, the axiom of choice, well-ordering, Zorn's lemma

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This chapter collects the set theory used in the rest of the course. Most of it is routine; the essential new tools are countability arguments, the axiom of choice, and its equivalent forms (well-ordering and Zorn's lemma).

1.1 Sets, functions and relations

We use the usual operations $A\cup B$, $A\cap B$, $A-B$, products $A\times B$, and De Morgan's laws $X-\bigcup A_\alpha = \bigcap(X-A_\alpha)$, $X-\bigcap A_\alpha = \bigcup(X-A_\alpha)$. A **function** $f:A\to B$ is a rule (formally a subset of $A\times B$) assigning to each $a\in A$ one $f(a)\in B$. It is **injective** if $f(a)=f(a')\Rightarrow a=a'$, **surjective** if $f(A)=B$, and **bijective** if both. Images and preimages satisfy

$$f^{-1}\Big(\bigcup B_\alpha\Big)=\bigcup f^{-1}(B_\alpha),\quad f^{-1}\Big(\bigcap B_\alpha\Big)=\bigcap f^{-1}(B_\alpha),\quad f^{-1}(B-C)=f^{-1}(B)-f^{-1}(C),$$

whereas images preserve unions but only $f(\bigcap A_\alpha)\subset\bigcap f(A_\alpha)$ in general (equality for injective f). Also $f(f^{-1}(B))\subset B$ and $A\subset f^{-1}(f(A))$.

A **relation** on A is a subset $C\subset A\times A$. An **equivalence relation** is reflexive, symmetric and transitive; its equivalence classes partition A, and conversely every partition comes from one. An **order relation** (simple or linear order) $<$ is comparable ($x\neq y\Rightarrow x<y$ or $y<x$), nonreflexive and transitive. Two ordered sets have the same **order type** if there is an order-preserving bijection between them. For subsets of an ordered set we have upper bounds, suprema and infima, and the **least upper bound property**: every nonempty subset bounded above has a supremum. $\mathbb{R}$ is characterized as an ordered field with this property; $\mathbb{Z}_+$ satisfies the **well-ordering property** (every nonempty subset has a least element), which is equivalent to induction.

1.2 Finite, countable and uncountable sets

A set is **finite** if it is empty or in bijection with $\{1,\dots,n\}$ for some n; this n (the cardinality) is unique, since there is no bijection of $\{1,\dots,n\}$ with a proper subset of itself (pigeonhole principle, by induction on n). A set is **countably infinite** if it is in bijection with $\mathbb{Z}_+$, and **countable** if finite or countably infinite.

Theorem 1.1

For nonempty B the following are equivalent: (a) B is countable; (b) there is a surjection $\mathbb{Z}_+\to B$; (c) there is an injection $B\to\mathbb{Z}_+$. Subsets of countable sets are countable; countable unions and finite products of countable sets are countable.

Proof. (c)⇒(a): an infinite subset $C\subset\mathbb{Z}_+$ is enumerated by $h(n)=$ the smallest element of $C-h(\{1,\dots,n-1\})$ (recursive definition, Section 1.3). (b)⇒(c): send b to the least element of its preimage. $\mathbb{Z}_+\times\mathbb{Z}_+\to\mathbb{Z}_+$, $(n,m)\mapsto 2^n3^m$ is injective. A countable union $\bigcup_n A_n$ is the image of $\mathbb{Z}_+\times\mathbb{Z}_+$ under $(n,m)\mapsto f_n(m)$, where $f_n:\mathbb{Z}_+\to A_n$ are surjections (chosen using the axiom of choice). ■

Theorem 1.2

$\{0,1\}^\omega$ (infinite binary sequences) and $\mathcal{P}(\mathbb{Z}_+)$ are uncountable. More generally, there is no surjection $A\to\mathcal{P}(A)$.

Proof. Given $g:A\to\mathcal{P}(A)$, the set $B=\{a:a\notin g(a)\}$ is not in the image: if $B=g(a_0)$, then $a_0\in B\iff a_0\notin B$. For sequences, the diagonal sequence $y_n=1-x_n(n)$ differs from every x_n. ■

So there are infinitely many sizes of infinite sets. The **Schröder–Bernstein theorem** states that injections $A \to B$ and $B \to A$ give a bijection; in particular $\mathbb{R}$, $\mathcal{P}(\mathbb{Z}_+)$ and $\{0,1\}^\omega$ have the same cardinality.

1.3 Recursive definition and the axiom of choice

Theorem 1.3 (Principle of recursive definition)

Given a set A, an element $a_1 \in A$, and a rule ρ assigning to each function $f : \{1,\dots,n\} \to A$ an element $\rho(f) \in A$, there is a unique $h : \mathbb{Z}_+ \to A$ with $h(1) = a_1$ and $h(n) = \rho(h|_{\{1,\dots,n-1\}})$ for $n > 1$.

Proof. Show by induction that for each n there is a unique function on $\{1,\dots,n\}$ satisfying the formula there; these functions are compatible, and their union is h. ∎

Definition (Axiom of choice)

For any collection $\mathcal{A}$ of disjoint nonempty sets there is a set containing exactly one element of each member of $\mathcal{A}$. Equivalently, for any collection $\mathcal{B}$ of nonempty sets there is a **choice function** $c : \mathcal{B} \to \bigcup \mathcal{B}$ with $c(B) \in B$.

Theorem 1.4

A set A is infinite iff there is an injection $\mathbb{Z}_+ \to A$ iff A is in bijection with a proper subset of itself.

Proof. If A is infinite, choose (with a choice function c on nonempty subsets of A) recursively $f(n) = c(A - f(\{1,\dots,n-1\}))$. Given an injection f, the map shifting $f(n) \mapsto f(n+1)$ and fixing the rest is a bijection onto a proper subset. A finite set has no such bijection. ∎

1.4 Well-ordered sets and Zorn's lemma

A set is **well-ordered** if it is simply ordered and every nonempty subset has a least element. Each element α of a well-ordered set J has a **section** $S_\alpha = \{x : x < \alpha\}$.

Theorem 1.5 (Well-ordering theorem, Zermelo)

Every set can be well-ordered. (This is equivalent to the axiom of choice.)

Proposition 1.6

There is a well-ordered set S_Ω that is uncountable but every section of which is countable; it is unique up to order type and is called the **minimal uncountable well-ordered set**. Every countable subset of S_Ω has an upper bound in S_Ω.

Proof. Well-order an uncountable set B, append a largest element, and let Ω be the least element whose section is uncountable; then S_Ω works. A countable subset A has $\bigcup_{a\in A} S_a$ countable, hence not all of S_Ω; any element outside is an upper bound. ∎

S_Ω and $\overline{S}_\Omega = S_\Omega \cup \{\Omega\}$ give many counterexamples in later chapters.

A **strict partial order** $\prec$ is nonreflexive and transitive (not every pair need be comparable).

Theorem 1.7 (Zorn's lemma)

Let A be partially ordered. If every simply ordered subset (chain) of A has an upper bound in A, then A has a maximal element.

Idea. Using a choice function, build by transfinite recursion along a well-ordering a chain that can no longer be extended; its upper bound is maximal. (Equivalently, the Hausdorff maximum principle: every partially ordered set has a maximal chain.) ■

Example 1.1

Zorn's lemma shows that every vector space has a basis (a maximal linearly independent set spans), that every ring with identity has a maximal ideal, and (Chapter 5) it is the engine behind the Tychonoff theorem.